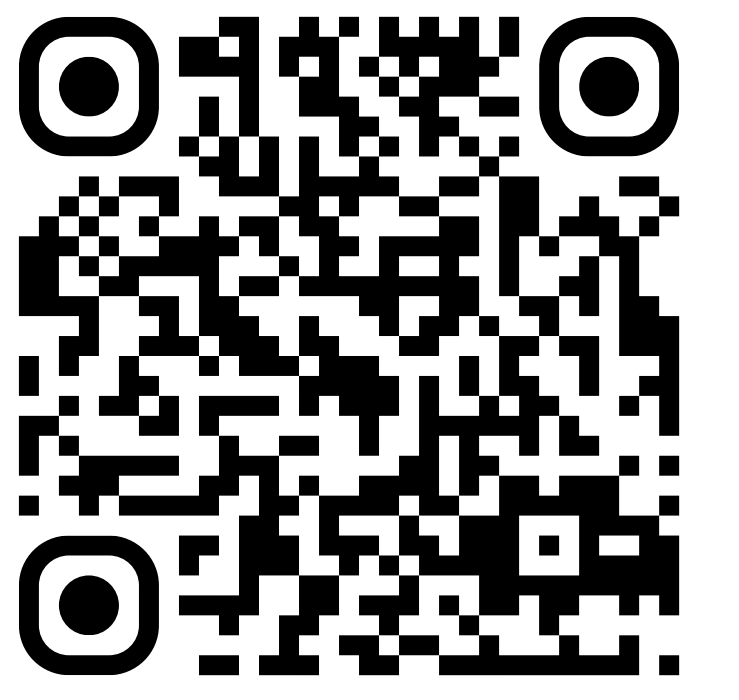


Rethinking Information Synthesis in Multimodal Question Answering A Multi-Agent Perspective

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Motivation

Baseline (CoT) Comparison. CoT misclassifies “bird” due to token frequency bias. MAMMQA’s modular design ensures factual grounding and interpretability.

Example 2: “According to the chart and text, which city had the highest rainfall?”

Input: A bar chart with rainfall data and a paragraph on weather patterns.

Answer: Singapore.

Example 1: “Which animal in the image is a mammal?”

Input: An image showing a frog, a dolphin, and a bird, with the caption “These animals live in different habitats.”

Answer: Dolphin.

Stage 1 – Expert Agents: Image agent identifies animal regions; text agent extracts relevant concepts (“different habitats”).

Stage 2 – Synthesizer: Fuses visual and textual evidence to reason over biological traits.

Stage 3 – Aggregator: Produces the reasoning trace: [Visual: Dolphin → live birth] + [Text: warm-blooded] → Dolphin is mammal.

Multimodal QA remains fundamentally hard because current models struggle to reliably integrate **Text**, **Tabular**, and **Visual** information.

Many MLLMs suffer from hallucinations, ungrounded reasoning, or over-reliance on **linguistic priors** rather than actual evidence.

Multimodal QA needs error-detection and **abstention** not just **confident guesses**.

Contributions

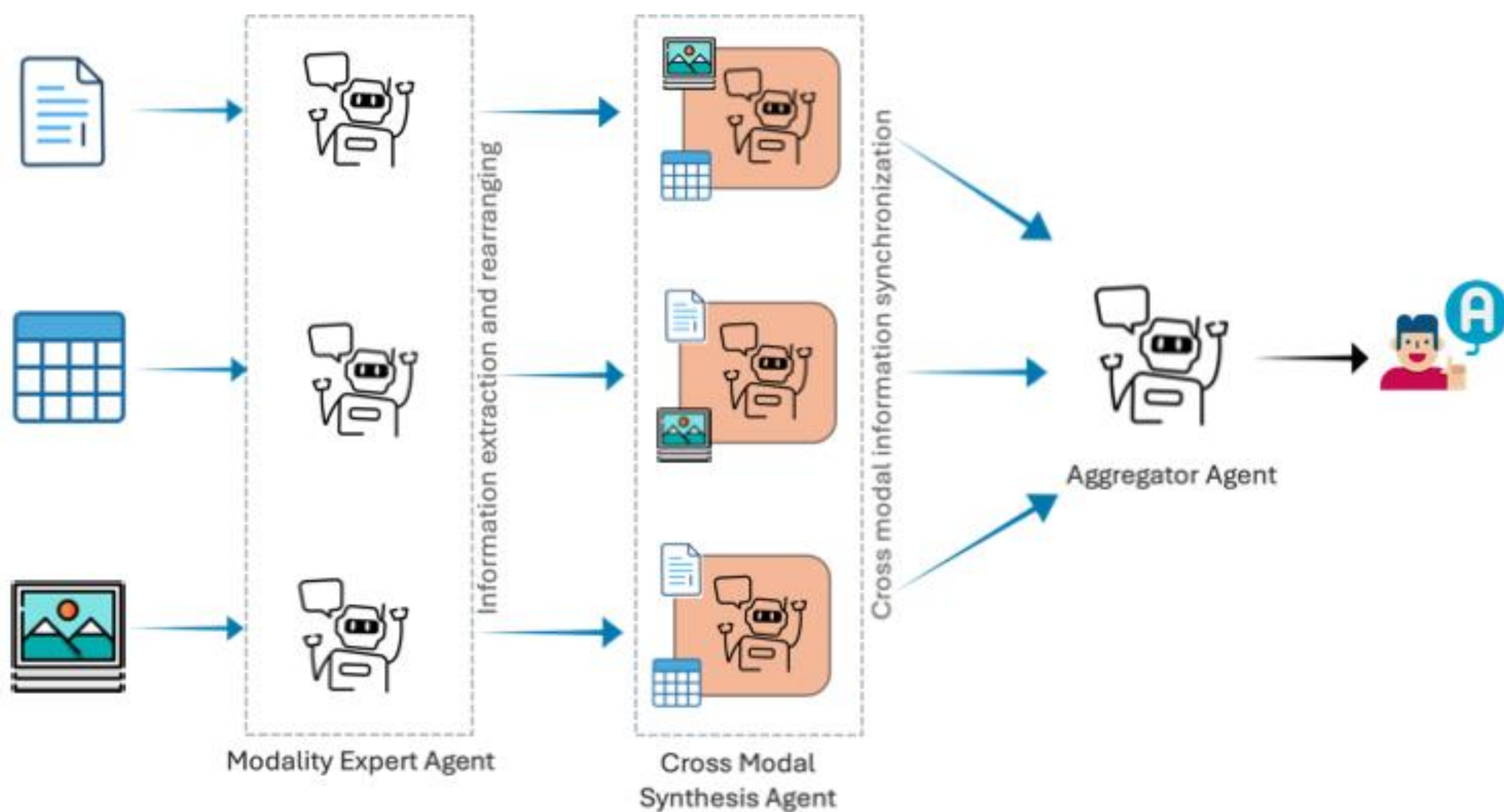
A fully prompt-driven multi-agent framework for multimodal QA

Unified and role-consistent agents across **Text**, **Tables**, and **Images**

Robustness, calibration, and reduced hallucination through evidence-based abstention

A transparent, modular, and scalable architecture that enables reliable and interpretable multimodal reasoning.

Overview of MAMMQA



- **Multi-Agent** framework for multimodal QA over **Text**, **Tables**, and **Images**
- **Three specialized agents:**
 1. **Modality Expert Agent** – extracts modality-specific insights.
 2. **Cross-Modal Synthesis Agent** – aligns and fuses evidence across modalities.
 3. **Aggregator Agent** – resolves conflicts and produces a grounded final answer.
- **Robust:** avoids hallucinations through evidence-based abstention.

MAMMQA with Efficient Agents

Modality	Img	Tb Img	Tb Txt	Tb	Txt Img	Txt	Total
<i>OpenAI 4o Mini</i>							
CoT	33.15	53.81	66.67	84.55	55.95	77.67	64.60
CapCoT	53.91	64.98	69.05	84.14	61.90	77.33	70.39
ToT	54.97	63.35	64.37	67.70	61.11	69.65	64.88
Ours	61.31	70.30	81.58	89.16	59.75	85.57	76.37
<i>Gemini 1.5-Flash 8B</i>							
CoT	47.41	53.38	58.88	74.73	46.43	72.82	62.16
CapCoT	47.84	50.02	55.87	74.88	39.29	72.42	60.66
ToT	36.93	43.06	52.32	53.72	33.33	70.61	53.10
Ours	51.23	54.12	57.42	83.69	42.86	79.47	65.84
<i>Qwen 2.5 VL 7B Instruct</i>							
CoT	29.11	32.58	30.66	38.75	17.86	38.28	33.84
CapCoT	48.10	53.94	60.56	71.52	41.67	71.31	61.54
ToT	55.90	47.82	52.50	60.83	41.64	64.44	57.12
Ours	50.74	55.88	63.68	81.35	53.26	80.51	67.56
<i>Qwen 2.5 VL 3B Instruct</i>							
CoT	11.86	23.71	22.14	32.25	14.29	25.52	23.15
CapCoT	48.10	42.08	47.08	64.94	39.29	65.04	53.98
ToT	42.01	43.65	48.40	52.57	33.74	66.51	52.91
Ours	33.73	43.10	45.33	62.29	35.52	67.73	52.12

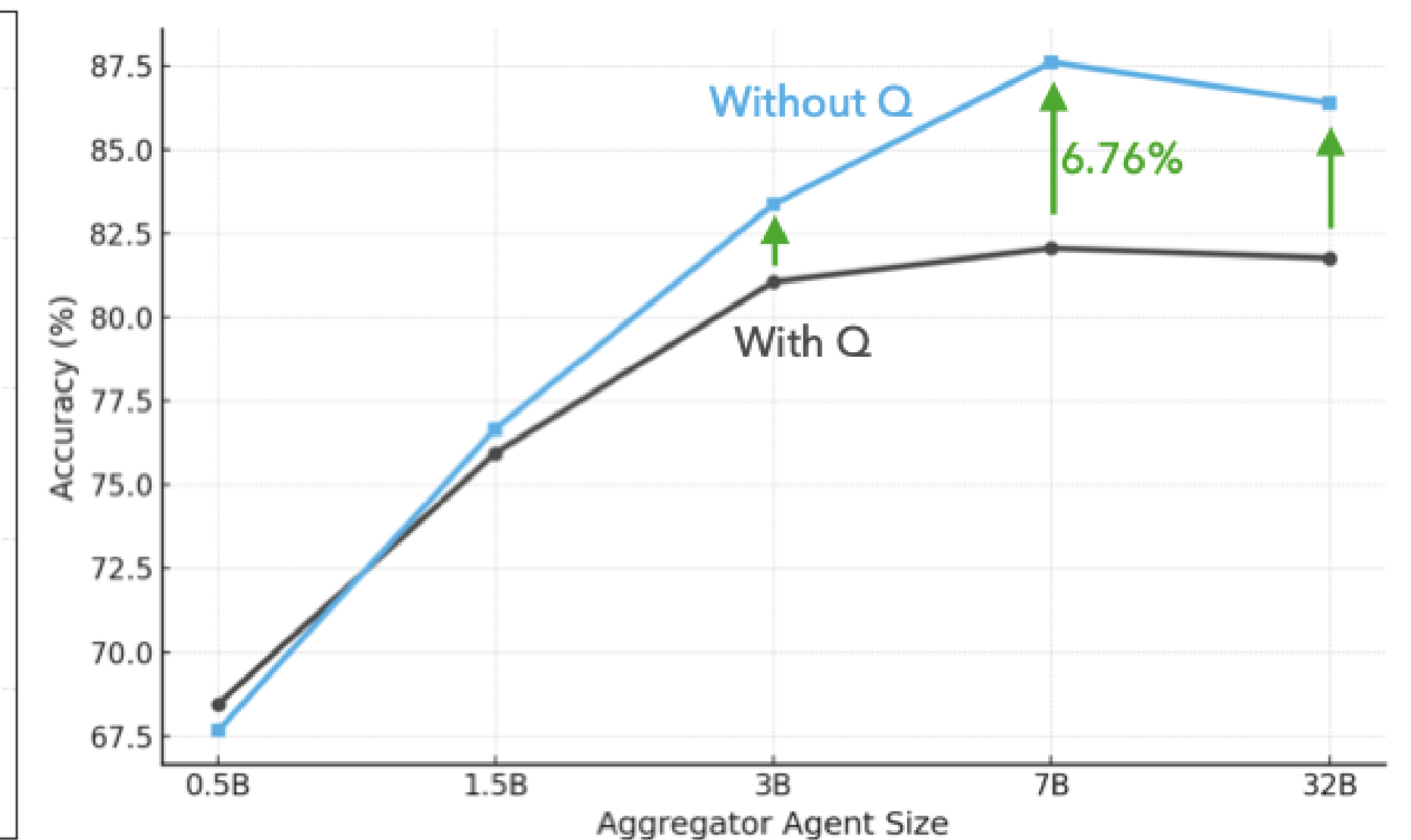
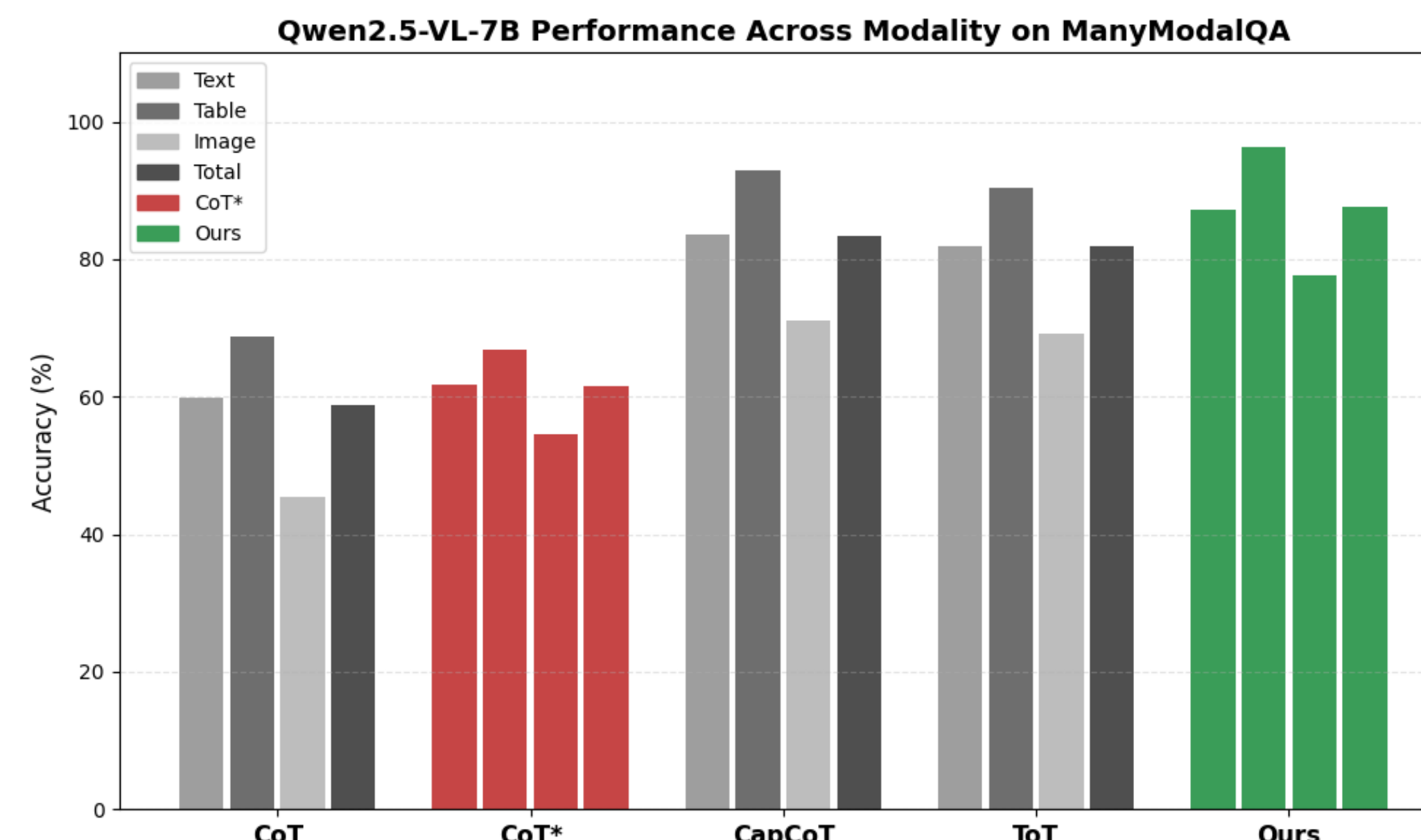
Table 1: Quantitative Analysis on MULTIMODALQA dataset

Superior Performance: Our method consistently outperforms CoT, ToT, and CapCoT across benchmarks (e.g., Qwen-7B: 87.61% vs. CoT 58.87%, ToT 81.89%, CapCoT 83.41%).

Caption Limitation: Caption-based reasoning fails when descriptions miss key details, making CapCoT unreliable.

Stronger Reasoning: Our multi-agent pipeline provides deeper, grounded reasoning by integrating visual, tabular, and textual evidence more faithfully.

Data Leakage in SOTA MLLMs



CoT* (no context) *should* perform worse but instead **outperforms** or **matches** its normal version on MANYMODALQA, exposing reliance on **memorized** priors rather than grounded evidence.

At 7B Aggregator, removing question improves accuracy by **+6.76%**

In MultiModalQA, models with CoT reproduce **typos** and outdated facts, indicating **memorization** in reasoning.